

Influence of free field variability on linear soil–structure interaction (SSI) by indirect integral representation

E. Savin^{*,†}

Laboratoire MSS-Mat, Ecole Centrale de Paris, 92295 Châtenay-Malabry Cedex, France

SUMMARY

An integral equation for the representation of the response of a structure impinged by an incident wave field including soil–structure interaction is proposed. It requires the knowledge of the fundamental solution for the overall soil–structure domain when a unit load is applied to the structure. This fundamental solution is obtained by means of a substructuring technique and boundary integral equations using the Green tensors for homogeneous or horizontally stratified soil media. The effects of a non-stationary modulated random incident field are addressed in terms of the instantaneous power spectral density of the structural response of interest for a given coherency function of the free field. Several applications of the proposed procedure are presented. The first one considers kinematic interaction of a rigid circular foundation and is used to validate the numerical implementation. The second one considers a complex structure on a stiff stratified soil and the last one considers the pounding effect between two adjacent, identical structures resting on a thin soft soil layer. Copyright © 2002 John Wiley & Sons, Ltd.

KEY WORDS: dynamic substructuring; boundary element method; spectral analysis; soil–structure interaction; pounding effect

1. INTRODUCTION

Recordings of strong motion arrays implemented on several test sites, such as SMART-1 in Taiwan (see for instance Reference [1]) or EUROSEISTEST in Greece [2], have indicated that seismic free fields exhibit some lateral variability even over short distances. This variability, apart from the purely deterministic wave passage effect, is strongly incoherent and can be attributed to various complex effects: the geology and stratification of the site, the wave propagation pattern, the wave components depending on the source mechanism, hysteretic damping and geometrical dispersion, diffusion, multiple scattering, etc. (see for instance the

* Correspondence to: E. Savin, ONERA-DDSS/MS, 29 Avenue de la Division Leclerc, 92322 Châtillon Cedex, France.

† E-mail: eric.savin@onera.fr

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recent works of Der Kiureghian [3], Zerva [4] and references therein). The relative importance of these phenomena is clearly dependent on the spatial scale of analysis and frequency range considered, however their intrinsic complexity justifies the use of a stochastic model for incident wave fields. In the instance of large spread footings, this spatial variability may induce significant differential displacements between two points, and then important additional displacements and efforts in the structure [5]. In the case of adjacent buildings, it is responsible for the pounding effect which has caused much irreversible damage observed after several recent earthquakes (see Anagnostopoulos [6] for a review). It is therefore desirable to develop adapted numerical methods to predict such phenomena accounting for the influence of linear soil–structure interaction; these methods should fall into the frame of random linear vibrations.

The analysis of random linear vibrations of elastic structures under stationary stochastic loads is classically based on a discretization by the finite element method (FEM) in order to compute a finite set of eigenmodes for the associated conservative system. The stationary response of the structure is obtained by standard spectral analyses which combine the contributions of a reduced number of degrees of freedom, namely the generalized co-ordinates on the truncated modal basis, when excitations are projected on this basis (see for instance Reference [7]). However this approach cannot be readily applied for vibration predictions in, for instance, the medium frequency range [8] or systems coupled with infinite media (see for instance Colton and Kress [9], Dautray and Lions [10]). Soil–structure interaction is an illustration of the latter limitation, in the sense that it can hardly be computed by the usual finite element method. Indeed it would need the consideration of absorbing conditions at the boundary of the mesh formulated to radiate both the body and surface waves of the Rayleigh, Love or even Stoneley-type typically arising in half-spaces and stratified media. The effects of corners and edges for such artificial boundaries should also be considered. Attempts have been made to write such conditions for elastic waves in highly heterogeneous media [11], but an efficient and robust numerical implementation remains problematical. Furthermore, it is worth noting that a proper characterization of the foundation impedance would usually require large meshes of the soil were a standard finite element method to be used, especially for the pumping mode [12, 13]. Boundary element methods (BEM) have proved to be much more efficient in such a case when they are implemented in conjunction with substructuring techniques [12, 14]. The radiation conditions are automatically taken into account by the Green kernels, which are, however, known only for particular soil profiles. The numerical method to calculate the Green function of a stratified half-space is now well established [15] and its solution accounts naturally for the body and guided waves. For more simple configurations, such as homogeneous media, it is however recognized that infinite elements are presumably more efficient than BEM from a computational point of view, as demonstrated by Burnett [16] amongst others. Analytical or semi-analytical methods [17, 18] are also often more appropriate to certain geometries.

To study random vibrations of a structure under stochastic incident fields, either trajectories of the excitation process can be simulated [19, 20], or spectral analysis techniques can be used [21]. Both approaches require the knowledge of some statistical properties of the incident waves on one hand, say their mean and auto-correlation functions if they are modelized by second-order processes, and the impulse or frequency response function of the coupled soil–structure system on the other hand. The main purpose of this paper is to propose an integral representation of the response function for the structure including linear interactions with the soil, and whose dimension is the number of loaded degrees of freedom of the discretized

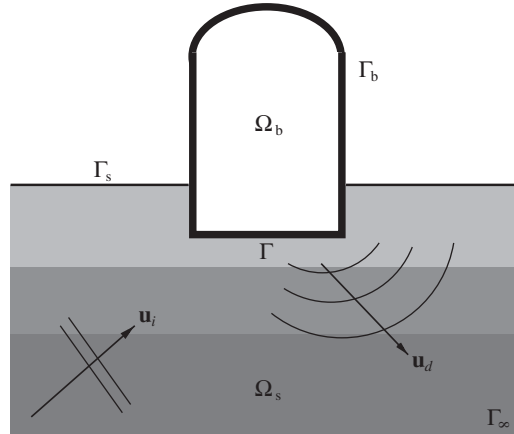


Figure 1. Model geometry of a structure occupying the bounded domain Ω_b coupled with a stratified soil occupying the half-space Ω_s and impinged by an incident wave $\mathbf{u}_i$.

soil–structure interface. Earthquake induced random vibrations of a structure and their variability have been considered by various authors, for instance for a multi-supported bridge [22] or for a rigid spread foundation whose displacement field is the random incident field averaged over its surface [23–25]. These approaches have neglected soil–foundation interactions. The works of Luco *et al.* [5, 26] or Païs and Kausel [27] have not, however they remain limited to kinematic interactions only. The present representation considers any quantity of interest for the structural response characterization, including dynamic interactions, for either superficial or grounded foundations.

This article is organized as follows. In Section 2 we summarize the basic notations as well as the main mechanical hypotheses used throughout the paper. Section 3 presents the main results, namely direct and indirect integral representations for the structural responses of interest. Then Section 4 describes how the spectral analysis theory of linear random vibrations may be applied to the indirect integral representation obtained in the previous section. In Section 5 we show how to compute the Green function of the overall system needed for the analysis, whereas numerical applications are presented in Section 6. We finally draw some conclusions in Section 7.

2. MODELLING OF THE COUPLED SOIL–STRUCTURE INTERACTION PROBLEM

2.1. Notations and general mechanical hypotheses

Let Ω_s be the half-space in $\mathbb{R}^3$ occupied by the soil, Ω_b be the bounded open domain occupied by the elastic structure, and let $\Omega = \Omega_s \cup \Omega_b$ be the overall domain. We consider throughout this paper linearized vibrations of the structure about a static equilibrium configuration, when coupled with the elastic soil medium (see Figure 1). Since a linear behaviour is assumed for both media, a frequency domain formulation can be used, the convention for the Fourier transform being $\mathbf{u}(\omega) = \int_{\mathbb{R}} e^{-i\omega t} \mathbf{u}(t) dt$ where ω denotes the circular frequency, $\mathbf{u}$ is any vector

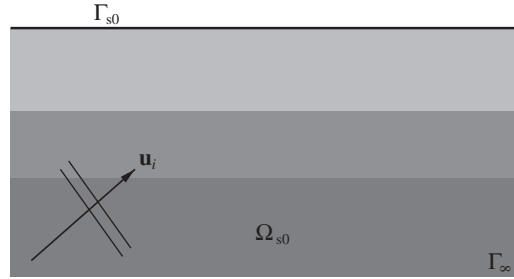


Figure 2. The soil half-space occupying Ω_{s0} before it is excavated for the construction of structural foundations and impinged by the incident wave $\mathbf{u}_i$.

in $\mathbb{C}^3$ and $\bar{\mathbf{u}}$ its conjugate ($\mathbb{R}$ and $\mathbb{C}$ are the sets of real and complex numbers, respectively). The usual scalar product of two vectors $\mathbf{u} = \{u_1, u_2, u_3\}$ and $\mathbf{v} = \{v_1, v_2, v_3\}$ is denoted by $\mathbf{u} \cdot \mathbf{v} = \sum_j u_j v_j$. A^T stands for the transpose of matrix A . We also use the following notations, for Ω_a being any bounded medium and $\Sigma \subseteq \partial\Omega_a$ its boundary or a part of it:

$$(\mathbf{u}, \mathbf{v})_a = \int_{\Omega_a} \mathbf{u} \cdot \mathbf{v} \, d\mathbf{x}$$

and:

$$\langle \mathbf{u}, \mathbf{v} \rangle_\Sigma = \int_\Sigma \mathbf{u} \cdot \mathbf{v} \, dS(\mathbf{x})$$

for two vectors $\mathbf{u}$ and $\mathbf{v}$.

Let $\Gamma = \partial\Omega_s \cap \partial\Omega_b$ be the interaction surface between Ω_b and Ω_s , and let Γ_∞ be formally the boundary of Ω_s at infinity. We denote by $\Gamma_b = \partial\Omega_b \setminus \Gamma$ the traction-free interface between the structure and the ambient air, and by Γ_s the traction-free surface of the soil half-space. The latter is impinged by an incident wave denoted by $\mathbf{u}_i(\mathbf{x}, \omega)$ and defined classically in the soil half-space occupying Ω_{s0} before it is excavated for the construction of structural foundations (see Figure 2). Let $\mathbf{u}(\mathbf{x}, \omega)$ be the total displacement field at each point $\mathbf{x}$ in Ω , then it satisfies the following homogeneous Navier equation:

$$\text{Div } \boldsymbol{\sigma}(\mathbf{u}) + \rho\omega^2 \mathbf{u} = \mathbf{0} \quad (1)$$

with boundary conditions $\mathbf{u} = \mathbf{u}_i$ on Γ_∞ and $\boldsymbol{\sigma}(\mathbf{u})\mathbf{n} = \mathbf{0}$ on $\Gamma_b \cup \Gamma_s$; $\rho^r(\mathbf{x})$ is the material mass density when $\mathbf{x} \in \Omega_r$, with $r = b$ or s (these superscripts are kept in the remainder of the paper to refer to Ω_b or Ω_s , respectively); $\boldsymbol{\sigma}(\mathbf{u})$ is the elastic stress tensor defined by $\boldsymbol{\sigma}^r(\mathbf{u}) = \mathbf{D}^r : \boldsymbol{\varepsilon}(\mathbf{u})$ where $\mathbf{D}^r(\mathbf{x}, \omega)$ is the fourth-order elasticity tensor of the materials when $\mathbf{x} \in \Omega_r$ and $\boldsymbol{\varepsilon}(\mathbf{u})$ is the linearized strain tensor. At last $\mathbf{n}^r$ denotes the outward unit normal vector to $\partial\Omega_r$ and the traction field on $\partial\Omega_r$ associated with the displacement field $\mathbf{u}$ is denoted by $\mathbf{t}^r(\mathbf{u}) = \boldsymbol{\sigma}^r(\mathbf{u})\mathbf{n}^r$. In Ω_b we will assume that the structure exhibits viscous damping, whereas in Ω_s we will assume that the soil medium exhibits hysteretic damping (independently of the frequency). The latter ensures that a homogeneous radiation condition can be imposed to the outgoing fields.

2.2. Hypothesis for the seismic free field

The seismic free field satisfies the homogeneous Navier equation in the soil half-space:

$$\text{Div } \boldsymbol{\sigma}^s(\mathbf{u}_i) + \rho^s \omega^2 \mathbf{u}_i = \mathbf{0} \quad (2)$$

with traction-free boundary condition on Γ_{s0} (see Figure 2): $\mathbf{t}^s(\mathbf{u}_i) = \mathbf{0}$. As already noted above, it represents the displacement field in the soil in the vicinity of the foundation due to an earthquake when it is assumed that the structure has not been built yet. The time varying free field $\mathbf{u}_i(\mathbf{x}, t)$ recorded on the soil–structure interface Γ is modelled by a second-order Gaussian non-homogeneous, non-stationary centred stochastic field $(\mathbf{U}_{\mathbf{x}, t}, \mathbf{x}, t \in \Gamma \times \mathbb{R}^+)$ with values in $\mathbb{R}^3$. It is therefore completely described by a 3×3 real matrix-valued auto-correlation function $\mathbf{R}_U$ defined on $\Gamma \times \Gamma \times \mathbb{R}^+ \times \mathbb{R}^+$ by

$$\mathbf{x}, \mathbf{y}, t, t' \mapsto \mathbf{R}_U(\mathbf{x}, \mathbf{y}, t, t') = \mathbb{E}\{\mathbf{U}_{\mathbf{x}, t} \otimes \mathbf{U}_{\mathbf{y}, t'}\} \quad (3)$$

where $\mathbb{E}$ stands for mathematical expectation. A commonly used assumption is to write the incident field in the form (see for instance Reference [28]):

$$\mathbf{u}_i(\mathbf{x}, t) = \varphi(t) \mathbf{v}_i(\mathbf{x}, t) \quad \forall \mathbf{x} \in \Gamma \quad (4)$$

where $t \mapsto \varphi(t)$ is a deterministic continuous modulation function defined on $\mathbb{R}^+$; $\mathbf{v}_i(\mathbf{x}, t)$ is modelled by a second-order Gaussian non-homogeneous, mean square stationary centred stochastic field $(\mathbf{V}_{\mathbf{x}, t}, \mathbf{x}, t \in \Gamma \times \mathbb{R})$ with values in $\mathbb{R}^3$ such that its auto-correlation function is:

$$\mathbf{R}_V(\mathbf{x}, \mathbf{y}, t - t') = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\omega(t-t')} \mathbf{S}_V(\mathbf{x}, \mathbf{y}, \omega) d\omega \quad (5)$$

with the 3×3 complex matrix-valued spectral density $\mathbf{S}_V$ defined on $\Gamma \times \Gamma \times \mathbb{R}$. It has members assumed of the form, for $j, k \in \{1, 2, 3\}$ two directions of the incident field:

$$[\mathbf{S}_V(\mathbf{x}, \mathbf{y}, \omega)]_{jk} = \sqrt{S_{0j}(\mathbf{x}, \omega) S_{0k}(\mathbf{y}, \omega)} \gamma_{jk}(\mathbf{x}, \mathbf{y}, \omega) \quad (6)$$

where $S_{0j}(\mathbf{x}, \omega)$ is the local power spectral density of the incident field along direction j at position $\mathbf{x}$; the previous relation defines the complex-valued transverse coherency function $\gamma_{jk}(\mathbf{x}, \mathbf{y}, \omega)$ defined on $\Gamma \times \Gamma \times \mathbb{R}$.

2.3. Diffracted field

The definition of the free field in the previous section leads to the following decomposition of the displacement field $\mathbf{u}^s$ in the soil domain:

$$\mathbf{u}^s(\mathbf{x}, \omega) = \mathbf{u}_i(\mathbf{x}, \omega) + \mathbf{u}_d(\mathbf{x}, \omega) \quad \forall \mathbf{x} \in \Omega_s, \quad \forall \omega \in \mathbb{R} \quad (7)$$

$\mathbf{u}_d$ is the diffracted field in the soil, which satisfy the homogeneous Navier equation (1) in Ω_s , the traction-free surface condition $\mathbf{t}^s(\mathbf{u}_d) = \mathbf{0}$ on Γ_s , and $\mathbf{u}_d = \mathbf{0}$ on Γ_∞ (that is homogeneous outgoing radiation condition at infinity). This latter condition, thanks to hysteretic damping, guarantees the existence and uniqueness of the diffracted field [29].

3. INTEGRAL REPRESENTATION OF THE STRUCTURAL RESPONSE

We focus on quantities directly relevant in a design process, such as structural displacements or accelerations at given significant points, and including the effects of soil–structure interaction. Therefore, we introduce an integral representation of the displacement $\mathbf{u}(\boldsymbol{\xi}) \cdot \mathbf{e}$ at a point $\boldsymbol{\xi}$ on the structure along the direction of $\mathbf{e}$ such that $\|\mathbf{e}\| = 1$ when the overall domain Ω is impinged by the incident wave field $\mathbf{u}_i$ as described previously.

For that purpose, let $\mathbf{u}^g(\mathbf{x}; \boldsymbol{\xi}, \mathbf{e})$ be the Green function of the overall domain Ω , that is the displacement field observed at point $\mathbf{x}$ when a unit load is applied at point $\boldsymbol{\xi}$ along $\mathbf{e}$ ($\mathbf{u}^g$, as well as all other fields, is also dependent on the circular frequency ω , however, we remove this dependence to lighten notations, except where otherwise specified). It satisfies the following Navier equation:

$$-\text{Div } \boldsymbol{\sigma}(\mathbf{u}^g) - \rho \omega^2 \mathbf{u}^g = \delta(\mathbf{x} - \boldsymbol{\xi}) \mathbf{e} \quad (8)$$

where δ stands for Dirac's measure. The Green function also satisfies the traction-free surface condition $\mathbf{t}(\mathbf{u}^g) = \mathbf{0}$ on $\Gamma_b \cup \Gamma_s$ and the outgoing radiation condition $\mathbf{u}^g = \mathbf{0}$ at infinity (on Γ_∞). Hence we have the following integral equation on Γ_∞ , that is written in the sense of a limit as $\|\mathbf{x} - \boldsymbol{\xi}\| \rightarrow +\infty$:

$$\langle \mathbf{u}^g, \mathbf{t}^s(\mathbf{u}_d) \rangle_{\Gamma_\infty} - \langle \mathbf{u}_d, \mathbf{t}^s(\mathbf{u}^g) \rangle_{\Gamma_\infty} = 0 \quad (9)$$

The construction of such a Green function is shown in Section 5.

3.1. Direct integral representation

Multiplying Equation (1) by $\mathbf{u}^g$ and Equation (8) by $\mathbf{u}$, then integrating over the ball Ω_R of radius R , which is assumed sufficiently large enough such that $(\Omega_b \cup \Gamma) \subset \Omega_R$, we get:

$$\mathbf{u}(\boldsymbol{\xi}) \cdot \mathbf{e} = (\text{Div } \boldsymbol{\sigma}(\mathbf{u}), \mathbf{u}^g)_R - (\text{Div } \boldsymbol{\sigma}(\mathbf{u}^g), \mathbf{u})_R \quad (10)$$

Integrating by parts, then using the traction-free surface condition on $\Gamma_b \cup \Gamma_s$, and letting finally $R \rightarrow +\infty$ we get:

$$\mathbf{u}(\boldsymbol{\xi}) \cdot \mathbf{e} = \langle \mathbf{t}^s(\mathbf{u}), \mathbf{u}^g \rangle_{\Gamma_\infty} - \langle \mathbf{t}^s(\mathbf{u}^g), \mathbf{u} \rangle_{\Gamma_\infty} \quad (11)$$

By Equation (9) and decomposition (7) of the displacement field in the soil domain, Equation (11) rewrites

$$\mathbf{u}(\boldsymbol{\xi}) \cdot \mathbf{e} = \langle \mathbf{t}^s(\mathbf{u}_i), \mathbf{u}^g \rangle_{\Gamma_\infty} - \langle \mathbf{t}^s(\mathbf{u}^g), \mathbf{u}_i \rangle_{\Gamma_\infty} \quad (12)$$

The incident wave field and the Green function both satisfy an homogeneous Navier equation in the soil domain if $\boldsymbol{\xi}$ belongs to the structure domain. For these fields, Maxwell–Betti's reciprocity theorem for an unbounded open domain leads to the following integral equation:

$$\langle \mathbf{u}^g, \mathbf{t}^s(\mathbf{u}_i) \rangle_{\Gamma_\infty} - \langle \mathbf{u}_i, \mathbf{t}^s(\mathbf{u}^g) \rangle_{\Gamma_\infty} + \langle \mathbf{u}^g, \mathbf{t}^s(\mathbf{u}_i) \rangle_\Gamma - \langle \mathbf{u}_i, \mathbf{t}^s(\mathbf{u}^g) \rangle_\Gamma = 0 \quad (13)$$

where again it has been taken account of the traction-free boundary condition on Γ_s . Using the above equation, we finally get a direct integral representation of the structural displacement at position $\boldsymbol{\xi}$ parallel to $\mathbf{e}$ as

$$\mathbf{u}(\boldsymbol{\xi}) \cdot \mathbf{e} = \langle \mathbf{t}^s(\mathbf{u}^g), \mathbf{u}_i \rangle_\Gamma - \langle \mathbf{t}^s(\mathbf{u}_i), \mathbf{u}^g \rangle_\Gamma \quad (14)$$

3.2. Indirect integral representation

The direct integral representation of the structural displacement of interest requires the calculation of the traction field on the soil–structure interface Γ associated to the free field. In a stochastic approach based on spectral analysis, it imposes the calculation of the auto-correlation function of the traction field $\mathbf{t}^s(\mathbf{U}_{\mathbf{x},t})$ and the cross-correlation function of $\mathbf{U}_{\mathbf{x},t}$ and $\mathbf{t}^s(\mathbf{U}_{\mathbf{x},t})$. This can be done formally by simple derivations, provided some regularity hypothesis for the auto-correlation function of $\mathbf{U}_{\mathbf{x},t}$. However, we show in this subsection that a much more efficient integral representation can be obtained, which does only depend on the free field on the interface. This approach is based on the explicit knowledge of the kernels of the single and double layer operators on Γ in the soil domain, these kernels being the first and second Green tensors $\mathbf{U}^s$ and $\mathbf{T}^s$ for the soil medium alone. If $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ is an orthonormal basis of $\mathbb{R}^3$, then $[\mathbf{U}^s(\boldsymbol{\xi}, \mathbf{x})]_{jk}$ and $[\mathbf{T}^s(\boldsymbol{\xi}, \mathbf{x})]_{jk}$ are, respectively, the displacement and traction fields observed in the soil along $\mathbf{e}_k$ at position $\mathbf{x}$ when a unit load is applied in it at $\boldsymbol{\xi}$ in the direction $\mathbf{e}_j$. They are computed analytically for homogeneous media, or numerically for stratified half-spaces [15].

The Green function $\mathbf{u}^g$ of the soil–structure coupled system occupying the overall domain Ω satisfies the homogeneous Navier equation in the unbounded soil medium Ω_s and the homogeneous outgoing radiation condition at infinity, so that it is the solution of the boundary integral equation:

$$\int_{\Gamma} \mathbf{U}^s(\mathbf{x}, \mathbf{y}) \mathbf{t}^s(\mathbf{u}^g)(\mathbf{y}) dS(\mathbf{y}) - \int_{\Gamma} \mathbf{T}^s(\mathbf{x}, \mathbf{y}) \mathbf{u}^g(\mathbf{y}) dS(\mathbf{y}) = \boldsymbol{\kappa}^s \mathbf{u}^g(\mathbf{x}), \quad \mathbf{x} \in \Gamma \quad (15)$$

where $\boldsymbol{\kappa}^s$ is the so-called free term operator defined by

$$\boldsymbol{\kappa}^s(\mathbf{x}) = \lim_{\varepsilon \rightarrow 0} \int_{S(\mathbf{x}, \varepsilon) \cap \Omega_s} \mathbf{T}^s(\mathbf{x}, \mathbf{y}) dS(\mathbf{y})$$

for $S(\mathbf{x}, \varepsilon)$ being the sphere centred at $\mathbf{x}$ with radius ε . The free field also satisfies the homogeneous Navier equation written in the soil domain actually occupied by the grounded foundations. Using the traction-free surface condition on Γ_{s0} , and since the outward normal to Γ seen from the soil volume actually occupied by the foundations is reversed, we have:

$$- \int_{\Gamma} \mathbf{U}^s(\mathbf{x}, \mathbf{y}) \mathbf{t}^s(\mathbf{u}_i)(\mathbf{y}) dS(\mathbf{y}) + \int_{\Gamma} \mathbf{T}^s(\mathbf{x}, \mathbf{y}) \mathbf{u}_i(\mathbf{y}) dS(\mathbf{y}) = (\mathbf{I} - \boldsymbol{\kappa}^s) \mathbf{u}_i(\mathbf{x}), \quad \mathbf{x} \in \Gamma \quad (16)$$

where $\mathbf{I}$ stands for identity.

Now let us consider the single layer elastic potential density $\mathbf{q}^g$ associated to the displacement field $\mathbf{u}^g$ and defined (uniquely) by

$$\mathbf{u}^g(\mathbf{x}; \boldsymbol{\xi}, \mathbf{e}) = \int_{\Gamma} \mathbf{U}^s(\mathbf{x}, \mathbf{y}) \mathbf{q}^g(\mathbf{y}; \boldsymbol{\xi}, \mathbf{e}) dS(\mathbf{y}) \quad (17)$$

That this integral equation (of the first kind) is uniquely invertible stems from the fact that, again, hysteretic damping has been considered in the soil, in such a way that wave numbers have a non-zero imaginary part. Therefore, there is no unicity deficiency at the so-called irregular frequencies of the interior Neumann problem (see Reference [9]). From this representation,

we obtain the associated traction field on Γ having in mind that the normal derivative of the single layer potential is discontinuous through Γ ([9, 10]):

$$\mathbf{t}^s(\mathbf{u}^g)(\mathbf{x}; \boldsymbol{\xi}, \mathbf{e}) = \int_{\Gamma} \mathbf{T}^{s*}(\mathbf{x}, \mathbf{y}) \mathbf{q}^g(\mathbf{y}; \boldsymbol{\xi}, \mathbf{e}) dS(\mathbf{y}) + \boldsymbol{\kappa}^s \mathbf{q}^g(\mathbf{x}; \boldsymbol{\xi}, \mathbf{e}), \quad \mathbf{x} \in \Gamma \quad (18)$$

where $\mathbf{T}^{s*}$ stands for the kernel of the adjoint of the integral operator (double layer potential) of kernel $\mathbf{T}^s$.

Remark 1

For a superficial foundation such that $\Gamma \subset \Gamma_{s0}$, we have $\boldsymbol{\kappa}^s \equiv \mathbf{I}$ [4] and $\mathbf{T}^s \equiv \mathbf{0}$ thus from Equation (18):

$$\mathbf{q}^g(\mathbf{x}; \boldsymbol{\xi}, \mathbf{e}) = \mathbf{t}^s(\mathbf{u}^g)(\mathbf{x}; \boldsymbol{\xi}, \mathbf{e}) \quad (19)$$

Multiplying Equation (18) by $\mathbf{u}_i$, and then integrating over Γ , we have:

$$\begin{aligned} \langle \mathbf{t}^s(\mathbf{u}^g), \mathbf{u}_i \rangle_{\Gamma} &= \langle \mathbf{T}^{s*} \mathbf{q}^g, \mathbf{u}_i \rangle_{\Gamma} + \langle \boldsymbol{\kappa}^s \mathbf{q}^g, \mathbf{u}_i \rangle_{\Gamma} \\ &= \langle \mathbf{q}^g, \mathbf{T}^s \mathbf{u}_i \rangle_{\Gamma} + \langle \mathbf{q}^g, \boldsymbol{\kappa}^s \mathbf{u}_i \rangle_{\Gamma} \end{aligned} \quad (20)$$

Note that in the above we have simplified the notations by using the same symbols $\mathbf{U}^s$ and $\mathbf{T}^s$ for the single and double layer potentials (operators), respectively, and their kernels (tensors). As we also have:

$$\begin{aligned} \langle \mathbf{t}^s(\mathbf{u}_i), \mathbf{u}^g \rangle_{\Gamma} &= \langle \mathbf{t}^s(\mathbf{u}_i), \mathbf{U}^s \mathbf{q}^g \rangle_{\Gamma} \\ &= \langle \mathbf{U}^s \mathbf{t}^s(\mathbf{u}_i), \mathbf{q}^g \rangle_{\Gamma} \end{aligned} \quad (21)$$

owing to symmetry of the single layer potential, we finally obtain with Equations (20) and (21):

$$\langle \mathbf{t}^s(\mathbf{u}^g), \mathbf{u}_i \rangle_{\Gamma} - \langle \mathbf{t}^s(\mathbf{u}_i), \mathbf{u}^g \rangle_{\Gamma} = \langle \mathbf{q}^g, \mathbf{T}^s \mathbf{u}_i - \mathbf{U}^s \mathbf{t}^s(\mathbf{u}_i) + \boldsymbol{\kappa}^s \mathbf{u}_i \rangle_{\Gamma} \quad (22)$$

The indirect integral representation of structural displacements at position $\boldsymbol{\xi}$ and along the direction of $\mathbf{e}$ is obtained injecting Equation (16) in Equation (22) above and reads:

$$\mathbf{u}(\boldsymbol{\xi}) \cdot \mathbf{e} = \langle \mathbf{q}^g, \mathbf{u}_i \rangle_{\Gamma} \quad (23)$$

This representation is readily useable in spectral analysis, as shown in Section 4.

4. SPECTRAL ANALYSIS FOR THE STRUCTURAL RESPONSE

The purpose of this section is to characterize the power spectral density (PSD) of the structural response when it is impinged by a random incident field described by its auto-correlation function or transverse power spectral density. A well-known property of the PSD is that it allows the computation of various relevant design quantities such as crossing rates or expected

maxima. However, it has to be kept in mind that the formula available to date to evaluate these quantities remain valid only for particular properties of the incident field.

The central theoretical point in spectral analysis of stochastic fields is to identify the linear convolution filter applied to the excitation process (the input), i.e. the free field in our case. It is given here by Equation (23) written back in the time domain:

$$\mathbf{u}(\xi, t) \cdot \mathbf{e} = \int_0^t \int_{\Gamma} \mathbf{q}^g(\mathbf{x}, t - \tau; \xi, \mathbf{e}) \cdot \mathbf{u}_i(\mathbf{x}, \tau) dS(\mathbf{x}) d\tau \quad (24)$$

invoking causality for the bounds of time integration. The design displacement $\mathbf{u}(\xi, t) \cdot \mathbf{e}$ is modeled by a stochastic process $(U_t, t \in \mathbb{R}^+)$ with values in $\mathbb{R}$. By linearity of Equation (24) with respect to the load $\mathbf{u}_i$, and given the properties of the latter described in Section 2.2, it is Gaussian, centred and non-stationary as well. It is assumed second-ordered without any physical restriction to our model. The random design displacement process $(U_t, t \in \mathbb{R}^+)$ is therefore completely described by its real-valued auto-correlation function R_U defined on $\mathbb{R}^+ \times \mathbb{R}^+$ and given straightforwardly by:

$$R_U(t, t') = \frac{1}{2\pi} \int_{\mathbb{R}} d\omega \int \int_{\Gamma \times \Gamma} \bar{\mathbf{T}}^T(\mathbf{x}, t, \omega) \mathbf{S}_v(\mathbf{x}, \mathbf{y}, \omega) \mathbf{T}(\mathbf{y}, t', \omega) dS(\mathbf{x}) dS(\mathbf{y}) \quad (25)$$

with the following expression for a pseudo-equivalent deterministic force applied to the foundation:

$$\mathbf{T}(\mathbf{x}, t, \omega) = \int_0^t e^{i\omega\tau} \varphi(\tau) \mathbf{q}^g(\mathbf{x}, t - \tau; \xi, \mathbf{e}) d\tau \quad (26)$$

and having values in $\mathbb{C}^3$. The above equation yields the following expression for the evolutionary power spectral density of the design displacement:

$$S_U(t, \omega) = \int \int_{\Gamma \times \Gamma} \bar{\mathbf{T}}^T(\mathbf{x}, t, \omega) \mathbf{S}_v(\mathbf{x}, \mathbf{y}, \omega) \mathbf{T}(\mathbf{y}, t, \omega) dS(\mathbf{x}) dS(\mathbf{y}) \quad (27)$$

which is defined on $\mathbb{R}^+ \times \mathbb{R}$ and has values in $\mathbb{R}$. It is easily verified that $\omega \mapsto S_U(t, \omega)$ is an even function for all $t \in \mathbb{R}^+$, as expected. The standard deviation of the design displacement can be computed by

$$\sigma_U^2(t) = 2 \int_0^{+\infty} S_U(t, \omega) d\omega \quad (28)$$

Remark 2

The integral representation (23) shows that knowledge of the incident field is required only on the soil–structure interface Γ , whether it is grounded or not. Applications of the previous results require therefore full knowledge of the second-order properties of the incident field in depth, whereas measurements have often been made only on the surface of the soil at a given site. This is the main limitation of the proposed methodology, since only little information is available for the in-depth statistical properties of seismic waves. Deconvolution may be used, as proposed by Kausel and Païs [30], to reconstruct in-depth transverse coherency functions from the estimated ones at the ground surface, using however several simplifying assumptions for the propagation patterns of the free field.

5. COMPUTATION OF THE GREEN FUNCTION OF THE OVERALL DOMAIN

Computation of structural displacements and their statistical properties requires knowledge of the Green function for the overall soil–structure coupled domain with unit loads applied to the structure (at the observed locations) only. Calculation of this Green function is a classical linear soil–structure interaction problem which can be solved by dynamic substructuring techniques using the fixed-interface eigenmodes of the structure [31, 32] and a reduced matrix model for its dynamic stiffness. The elastodynamic fields in the unbounded soil medium are obtained by boundary integral equations of the kind of Equation (15) and their regularized version for stratified media [33]. This procedure has been applied to many practical engineering problems such as seismic soil–structure interaction with random soil parameters, arch dam analysis under earthquake, seismic site effects for a complex topography, seismic borehole modelling, vibrations induced in the soil by moving vehicles, waves in periodic heterogeneous media [14, 34–36]. A detailed presentation of the procedure outlined below can be found for instance in Reference [36]. It is essentially a hybrid FEM/BEM approach [12] in as much as BEM is used to compute the Green function of the overall domain in the soil while FEM is used to evaluate it in the structure through the computation of its fixed-interface eigenmodes. As such, a FEM mesh could be used as well to discretize deterministic or random heterogeneities of the soil near foundations [36] if it deviates from a perfectly horizontal stratification. The remaining part of the stratified soil in this case would still be modelized by the BEM. The substructuring approach detailed below allows to couple the BEM and FEM.

5.1. Decomposition of the Green function in each subdomain

Substructuring starts from the equilibrium of the interface between the structure and the soil written in the weak sense in terms of the trace of the Green function $\mathbf{u}^g$ on Γ . The latter is expanded on a given basis $\{\phi_m\}_{1 \leq m \leq M}$ of the admissible displacement fields on Γ :

$$\mathbf{u}^g(\mathbf{x}; \xi, \mathbf{e}) = \sum_{m=1}^M c_m(\xi, \mathbf{e}) \phi_m(\mathbf{x}) \quad (29)$$

The first M_A fixed-interface eigenmodes of the structure are denoted by $\{\phi_A\}_{1 \leq A \leq M_A}$ with corresponding eigenvalues $\{\omega_A\}_{1 \leq A \leq M_A}$ and critical viscous damping rates $\{\zeta_A\}_{1 \leq A \leq M_A}$. The elastostatic lifting operator in the structure being denoted by $\mathcal{S}^b$, the elastostatic fields $\{\mathcal{S}^b \phi_m\}_{1 \leq m \leq M}$ are the so-called static modes, or constraint modes. Then the Green function is written as

$$\mathbf{u}^g(\mathbf{x}; \xi, \mathbf{e}) = \sum_{m=1}^M c_m(\xi, \mathbf{e}) \mathcal{S}^b \phi_m(\mathbf{x}) + \sum_{A=1}^{M_A} \alpha_A(\xi, \mathbf{e}) \phi_A(\mathbf{x}) \quad (30)$$

in the structure ($\mathbf{x} \in \Omega_b$), and

$$\mathbf{u}^g(\mathbf{x}; \xi, \mathbf{e}) = \sum_{m=1}^M c_m(\xi, \mathbf{e}) \mathbf{u}_m^s(\mathbf{x}) \quad (31)$$

in the soil ($\mathbf{x} \in \Omega_s$). In the above $\{\mathbf{u}_m^s\}_{1 \leq m \leq M}$ are the elementary scattered fields in Ω_s when boundary conditions $\mathbf{u}_m^s = \phi_m$ hold on Γ . They satisfy the homogeneous Navier equation in

Ω_s and the homogeneous outgoing radiation condition at infinity. Thus, they can be computed by representation integral equations:

$$\mathbf{u}_m^s(\mathbf{x}) = \int_{\Gamma} \mathbf{U}^s(\mathbf{x}, \mathbf{y}) \mathbf{t}_m^s(\mathbf{y}) dS(\mathbf{y}) - \int_{\Gamma} \mathbf{T}^s(\mathbf{x}, \mathbf{y}) \phi_m(\mathbf{y}) dS(\mathbf{y}), \quad \mathbf{x} \in \Omega_s, \quad m = 1, \dots, M \quad (32)$$

where traction fields $\{\mathbf{t}_m^s\}_{1 \leq m \leq M}$ on Γ are solutions of the boundary integral equations:

$$\int_{\Gamma} \mathbf{U}^s(\mathbf{x}, \mathbf{y}) \mathbf{t}_m^s(\mathbf{y}) dS(\mathbf{y}) = \boldsymbol{\kappa}^s \phi_m(\mathbf{x}) + \int_{\Gamma} \mathbf{T}^s(\mathbf{x}, \mathbf{y}) \phi_m(\mathbf{y}) dS(\mathbf{y}), \quad \mathbf{x} \in \Gamma, \quad m = 1, \dots, M \quad (33)$$

Note that $\boldsymbol{\kappa}^s(\mathbf{x}) = \frac{1}{2} \mathbf{I}$ if $\mathbf{x}$ lies within an homogeneous medium and Γ is regular at this point, but this simple, well-known result does not hold, as a general rule, at the interface between two layers. A regularized version of Equation (33) [33] is used which allows to circumvent this difficulty.

5.2. Coupled problem

The generalized co-ordinates $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_{M_A})^T$ and $\mathbf{c} = (c_1, \dots, c_M)^T$ are shown to be solution of the algebraic system:

$$\begin{bmatrix} \mathcal{Z}^b(\omega) & \mathcal{A}(\omega) \\ \mathcal{A}(\omega)^T & Z_{\Gamma}^b(\omega) + Z_{\Gamma}^s(\omega) \end{bmatrix} \begin{pmatrix} \boldsymbol{\alpha} \\ \mathbf{c} \end{pmatrix} = \begin{pmatrix} \mathbf{F}^{\Phi} \\ \mathbf{F}^{\Phi} \end{pmatrix} \quad (34)$$

where $Z_{\Gamma}^b(\omega)$ is the Guyan static condensation on Γ of the mass, stiffness [37] and damping matrices of the structure constructed using the elastostatic lifting operator $\mathcal{S}^b$; $Z_{\Gamma}^s(\omega)$ is the impedance matrix of the soil seen from Γ whose elements are:

$$[Z_{\Gamma}^s(\omega)]_{mn} = \langle \mathbf{t}_n^s, \phi_m \rangle_{\Gamma}$$

$\mathcal{Z}^b(\omega)$ is the dynamic stiffness matrix of the structure projected on its eigenmodes, which is diagonal assuming that they diagonalize the damping matrix as well:

$$\mathcal{Z}^b(\omega) = \text{diag}(-\omega^2 + 2i\zeta_A \omega_A \omega + \omega_A^2)$$

$\mathcal{A}(\omega)$ is the matrix of the classical participation factors whose elements are:

$$[\mathcal{A}(\omega)]_{Am} = -\omega^2(\rho^b \phi_A, \mathcal{S}^b \phi_m)_b$$

Lastly the forcing vectors have elements $F_A^{\Phi} = \phi_A(\xi) \cdot \mathbf{e}$ and $F_m^{\Phi} = \mathcal{S}^b \phi_m(\xi) \cdot \mathbf{e}$, respectively. $\mathcal{Z}^b(\omega)$, $Z_{\Gamma}^b(\omega)$ and $\mathcal{A}(\omega)$ may be evaluated numerically by the FEM, for instance. Once Equation (34) has been solved for $\boldsymbol{\alpha}$ and $\mathbf{c}$, the Green function in Ω_s is reconstructed by Equation (31) and then $\mathbf{q}^g$ is obtained by Equation (17). In particular, we have for the case of superficial foundations (see Equation (19)):

$$\mathbf{q}^g(\mathbf{x}; \xi, \mathbf{e}) = \sum_{m=1}^M c_m(\xi, \mathbf{e}) \mathbf{t}_m^s(\mathbf{x}) \quad (35)$$

6. NUMERICAL APPLICATIONS

The previous developments have been implemented in the computer code MISS-3D [38] dedicated to multi-domains and solid–fluid–structure interaction analyses by dynamic substructuring, boundary integral equations and the boundary element method using analytical and numerical Green functions. For the reasons mentioned in Remark 2, we consider only superficial foundations in the following examples, although the code developed so far considers any kind of grounded or superficial foundations. The first example presented below deals with kinematic interaction for a rigid, massless foundation over a homogeneous half-space and compares the results obtained with others published in the literature. The second one deals with a complex structure over a stratified soil including as much as four layers. The last example deals with the pounding effect between two nearby structures resting on a thin soil layer. These simulations consider also various models of transverse coherency function for the free field.

6.1. Kinematic interaction of a rigid, massless foundation

The main purpose of this example is to validate the proposed methodology by comparisons with the tabulated results given by Luco and Mita [26]. We consider only kinematic interaction, such that the generalized co-ordinates $\mathbf{c}^{\text{kin}}$ are the solution of Equation (34) where the contribution of structural eigenmodes is ignored:

$$[\mathbf{Z}_\Gamma^s(\omega)]\mathbf{c}^{\text{kin}} = \mathbf{F}^\Phi \quad (36)$$

We focus on the stationary response induced by a stochastic incident field $(\mathbf{V}_{\mathbf{x},t}, \mathbf{x}, t \in \Gamma \times \mathbb{R})$ propagating in a homogeneous soil half-space. The spectral density matrix of the kinematic generalized co-ordinates is easily shown, from the previous results, to have elements:

$$S_{mn}^{\text{kin}}(\omega) = \sum_{j=1}^3 \sum_{k=1}^3 \sqrt{S_{0j}(\omega)S_{0k}(\omega)} A_{jk}^{mn}(\omega) \quad (37)$$

assuming that S_{0j} is independent of $\mathbf{x}$ for $j = 1, 2, 3$, and where:

$$A_{jk}^{mn}(\omega) = \iint_{\Gamma \times \Gamma} \mathbf{t}_m^s(\mathbf{x}, \omega) \cdot \mathbf{e}_j \overline{\mathbf{t}_n^s(\mathbf{y}, \omega) \cdot \mathbf{e}_k} \gamma_{jk}(\mathbf{x}, \mathbf{y}, \omega) dS(\mathbf{x}) dS(\mathbf{y}) \quad (38)$$

For a rigid foundation, the interface modes $\{\phi_m\}_{1 \leq m \leq M}$ are the rigid-body displacements ($M = 6$ for 3 translations and 3 rotations). The so-called covariance coefficients $\omega \mapsto A_{jk}^{mn}(\omega)$ have been tabulated in Reference [26] for various values of $j, k = 1, 2, 3$ and $m, n = 1, 2, \dots, 6$, and a transverse coherency function given by

$$\gamma_{jk}(\mathbf{x}, \mathbf{y}, \omega) = \exp \left[- \left(\frac{\eta \omega \|\mathbf{x} - \mathbf{y}\|}{c^s} \right)^2 \right] \quad (39)$$

where η is a (dimensionless) coherency parameter and c^s is the typical velocity of waves in the soil medium. Figure 3 displays the plots of $\sqrt{A_{11}^{11}}$, $\sqrt{A_{11}^{66}}$, $\sqrt{A_{33}^{44}}$ and $\sqrt{|\Re A_{31}^{46}|}$ ($\Re$ stands for the real part) as functions of the frequency number $a_0 = \omega R / c^s$ for a circular foundation of radius R . Lines are for the coefficients calculated with the code developed, and

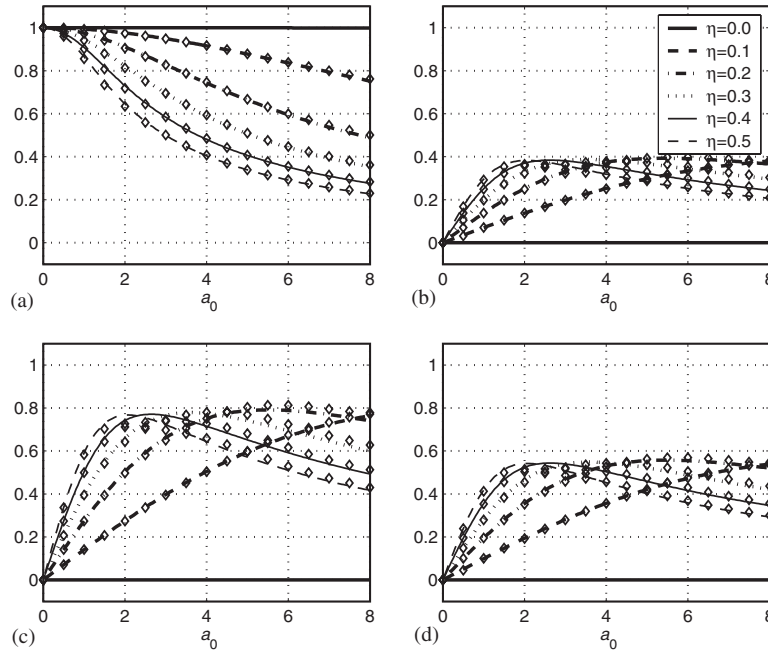


Figure 3. Covariance coefficients as functions of the frequency number a_0 calculated by the proposed methodology (lines) and as tabulated in Reference [26] (diamonds $\diamond$): (a) $a_0 \mapsto \sqrt{A_{11}^{11}(a_0)}$, (b) $a_0 \mapsto \sqrt{A_{11}^{66}(a_0)}$, (c) $a_0 \mapsto \sqrt{A_{33}^{44}(a_0)}$ and (d) $a_0 \mapsto \sqrt{|\operatorname{Re} A_{31}^{46}(a_0)|}$.

diamonds $\diamond$ point to the results tabulated in Reference [26] (where they have been calculated by simplified analytical formulas). The agreement is excellent and thus validates our numerical implementation.

6.2. Random vibrations of a complex structure

We consider in this subsection random vibrations of a complex structure (a nuclear power plant) resting on a circular spread footing of radius $R \simeq 25$ m. It is built on a stratified soil medium having four layers (including the bedrock) whose characteristics are summarized in Table I. This realistic example has already been considered in Reference [36] to study the influence of random soil parameters on the structural response under earthquake.

Structural displacements are expanded on the first $M_A = 30$ eigenmodes (corresponding to frequencies up to about 30 Hz) for a fixed-basis boundary condition. The fundamental frequency is $f_1 = 4.6$ Hz with multiplicity 2 for flexural motion with respect to the two horizontal principal axis of this almost symmetric structure. Damping ratios ζ_A have been set to 7% for all modes in the frequency range considered. Materials in each soil layer are assumed isotropic and as such are characterized by their Lamé's coefficients $\lambda^s(z)$ and $\mu^s(z)$ which depend on the depth z only. We consider, as in the previous example, stationary responses of the structure solicited by the incident wave process $(\mathbf{V}_{\mathbf{x},t}, \mathbf{x}, t \in \Gamma \times \mathbb{R})$ such that S_{0j} is again independent of

Table I. Characteristics of the stratified soil: thickness, Lamé's moduli and mass density (the hysteretic damping rate is 5%).

	h (m)	λ^s (MPa)	μ^s (MPa)	ρ^s (kg/m ³)
Layer #1	10.5	1.16×10^4	0.37×10^4	2400
Layer #2	10.5	1.95×10^4	0.76×10^4	2500
Layer #3	19.5	2.12×10^4	0.60×10^4	2400
Bedrock	∞	1.62×10^4	0.99×10^4	2500

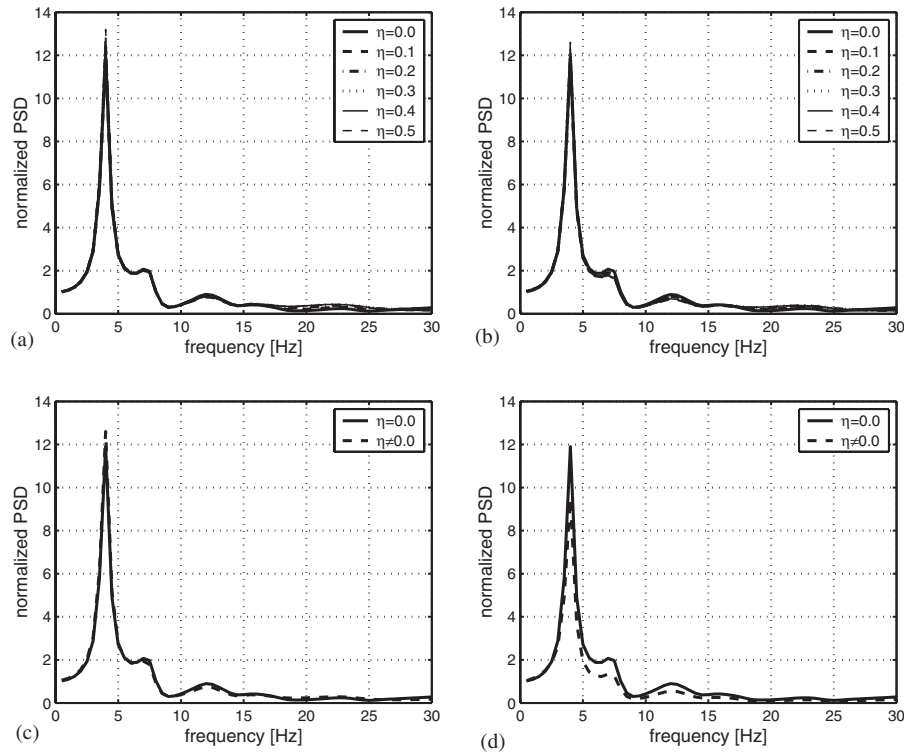


Figure 4. Normalized power spectral density of the horizontal displacement on top of a complex structure for different models of transverse coherency functions: (a) model of Luco and Wong [5], (b) model of Der Kiureghian [3], (c) model of Harichandran and Vanmarcke [39] and (d) model of Abrahamson *et al.* [40].

the position for $j = 1, 2, 3$. Figures 4 and 5 plot the power spectral density of horizontal displacements ($j = 1$) on top of the structure and on top of the foundation, respectively, for four different models of transverse coherency functions and normalized by $S_{01}(\omega)$. These coherency functions are, for $j, k = 1, 2, 3$: (a) the model of Luco and Wong [5], see Equation (39);

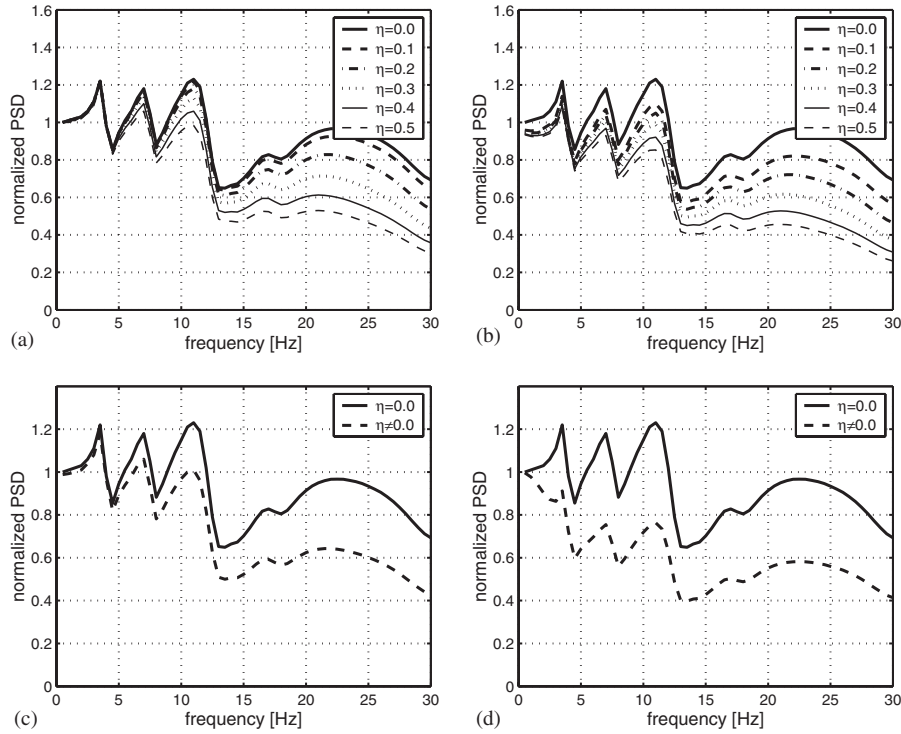


Figure 5. Normalized power spectral density of the horizontal displacement on top of the foundation of a complex structure for different models of transverse coherency functions: (a) model of Luco and Wong [5], (b) model of Der Kiureghian [3], (c) model of Harichandran and Vanmarcke [39] and (d) model of Abrahamson *et al.* [40].

(b) a model derived from the paper of Der Kiureghian [3] which is

$$\gamma_{jk}(\mathbf{x}, \mathbf{y}, \omega) = e^{-d^2} \cos\left(\frac{\pi}{2} \frac{d^\alpha}{1 + d^\alpha}\right) \quad (40)$$

with $d = \eta\omega\|\mathbf{x} - \mathbf{y}\|/c^s$ and $\alpha = 0.3$; (c) the model of Harichandran and Vanmarcke [39]:

$$\gamma_{jk}(\mathbf{x}, \mathbf{y}, \omega) = a \exp\left[\frac{-2\|\mathbf{x} - \mathbf{y}\|}{\alpha\beta(\omega)}(1 - a + \alpha a)\right] + (1 - a) \exp\left[\frac{-2\|\mathbf{x} - \mathbf{y}\|}{\beta(\omega)}(1 - a + \alpha a)\right] \quad (41)$$

where

$$\beta(\omega) = K \left[1 + \left(\frac{\omega}{\omega_0} \right)^b \right]^{-1/2}$$

with $a = 0.626$, $\alpha = 0.022$, $K = 19\,700\text{ m}$, $\omega_0 = 2\pi \times 2.02\text{ rad/s}$ and $b = 3.47$; and (d) the model of Abrahamson *et al.* [40]:

$$\tanh^{-1} |\gamma_{jk}(\mathbf{x}, \mathbf{y}, \omega)| = (a_1 + a_2 \|\mathbf{x} - \mathbf{y}\|) \left[\exp((b_1 + b_2 \|\mathbf{x} - \mathbf{y}\|)\omega) + \frac{1}{3}\omega^c \right] + K \quad (42)$$

with $a_1 = 2.54$, $a_2 = -0.012$, $b_1 = -0.115$, $b_2 = 0.00084$, $c = -0.878$ and $K = 0.35$. For these last two models, the case $\eta = 0$ corresponds to strictly coherent incident seismic waves, whereas the case $\eta \neq 0$ corresponds to incoherent waves (with coherency functions as above). We see from the plots that although the foundation movement is strongly dependent on the incoherence of the free field, especially at higher frequencies, the structure filters out this effect and the top response is almost independent of this incoherence, except for the model of Abrahamson *et al.* [40]. We also observe that the foundation horizontal movement is slightly decreased by the consideration of free field variability, in accordance with the conclusion reached by Luco and Wong [5]. Indeed it can be seen from Figure 3(a) that incoherent horizontal ground motions (along $\mathbf{e}_1$, $j = k = 1$) are filtered by a rigid foundation, observing its horizontal movement parallel to the same direction ($m = n = 1$). However, rotations about its normal axis (torsion) are increased when the free field incoherence is considered, whereas they should remain null for perfectly coherent incident waves polarized along $\mathbf{e}_1$ (see Figure 3(b)). Torsion rotations induced by the free field (horizontal) incoherence have not been observed for the present complex structure since its torsion eigenfrequencies lie outside the frequency range considered. It is worthy of note that for the present case soil–structure interaction could be ignored in a first approximation since the soil is very stiff compared to the structure (eigenfrequencies of the structure with and without coupling are almost unchanged). The next example considers stiff structures on a soft soil layer.

6.3. Pounding effect

The pounding effect has been investigated by various authors (see for example the recent works of Hao and Zhang [41], Behnamfar and Sugimura [42], Lin and Weng [43] and references therein), however soil–structure interaction has often been neglected. We consider here the case of two adjacent, *identical* buildings resting on rigid foundations and $\varepsilon = H/100$ apart, where H is the height of the buildings; then we compare ε with the standard deviation of differential displacements between them for incoherent incident seismic waves. The differential displacement between the first and second structures, at locations ξ_1 and ξ_2 respectively, parallel to the direction $\mathbf{e}$, is

$$\Delta(t) = [\mathbf{u}(\xi_1, t) - \mathbf{u}(\xi_2, t)] \cdot \mathbf{e} = \langle \mathbf{q}_1^g, \mathbf{u}_i \rangle_{\Gamma_1} - \langle \mathbf{q}_2^g, \mathbf{u}_i \rangle_{\Gamma_2} \quad (43)$$

where $\mathbf{q}_r^g$, $r = 1, 2$, are the single layer elastic potential densities for both structures considering the Green function of the coupled system occupying the overall domain $\Omega = \Omega_{b_1} \cup \Omega_{b_2} \cup \Omega_s$. If $\mathbf{u}_i(\mathbf{x}, t)$ is modellized by a stochastic process $(\mathbf{U}_{\mathbf{x}, t}, \mathbf{x}, t \in \Gamma \times \mathbb{R}^+)$ having the properties listed in Section 2.2, then $\Delta(t)$ can be modellized by a second-order, real-valued stochastic process $(\Delta_t, t \in \mathbb{R}^+)$ which is centred, Gaussian, non-homogeneous, non-stationary, with an

Table II. Mechanical characteristics of buildings and soil for the example of pounding analysis.

Buildings	$\rho^b = 2400 \text{ kg/m}^3$		
	ζ_A	Eigenfrequency (Hz)	
Flexural mode #1	2%	10.8	
Compressional mode #2	2%	31.2	
Soil	$\rho^s = 2420 \text{ kg/m}^3$		Hysteretic damping 2%
	$h \text{ (m)}$	$\lambda^s \text{ (MPa)}$	$\mu^s \text{ (MPa)}$
Layer	7	38.07×10^2	2.43×10^2
Bedrock	∞	85.93×10^2	5.48×10^2

instantaneous power spectral density given by

$$\begin{aligned}
S_{\Delta}(t, \omega) = & \int \int_{\Gamma_1 \times \Gamma_1} \bar{\mathbf{T}}_1^T(\mathbf{x}, t, \omega) \mathbf{S}_v(\mathbf{x}, \mathbf{y}, \omega) \mathbf{T}_1(\mathbf{y}, t, \omega) dS(\mathbf{x}) dS(\mathbf{y}) \\
& + \int \int_{\Gamma_2 \times \Gamma_2} \bar{\mathbf{T}}_2^T(\mathbf{x}, t, \omega) \mathbf{S}_v(\mathbf{x}, \mathbf{y}, \omega) \mathbf{T}_2(\mathbf{y}, t, \omega) dS(\mathbf{x}) dS(\mathbf{y}) \\
& - \int \int_{\Gamma_1 \times \Gamma_2} \bar{\mathbf{T}}_1^T(\mathbf{x}, t, \omega) \mathbf{S}_v(\mathbf{x}, \mathbf{y}, \omega) \mathbf{T}_2(\mathbf{y}, t, \omega) dS(\mathbf{x}) dS(\mathbf{y}) \\
& - \int \int_{\Gamma_2 \times \Gamma_1} \bar{\mathbf{T}}_2^T(\mathbf{x}, t, \omega) \mathbf{S}_v(\mathbf{x}, \mathbf{y}, \omega) \mathbf{T}_1(\mathbf{y}, t, \omega) dS(\mathbf{x}) dS(\mathbf{y}) \quad (44)
\end{aligned}$$

The standard deviation of differential displacements is computed by

$$\sigma_{\Delta}^2(t) = 2 \int_0^{+\infty} S_{\Delta}(t, \omega) d\omega \quad (45)$$

Mechanical characteristics of the soil and the structures studied in this example are summarized in Table II. Buildings displacements are expanded on the first two eigenmodes ($M_A = 2$ for both structures), the first one being a flexural mode about their horizontal axis (structures are axisymmetric) and the second one being a compressional mode about their vertical axis. The soil has one thin layer over an infinitely deep bedrock. Stochastic process $(\mathbf{U}_{\mathbf{x}, t}, \mathbf{x}, t \in \Gamma \times \mathbb{R}^+)$ is characterized by the transverse coherency function of Luco and Wong [5], local power spectral densities S_{0j} , $j = 1, 2, 3$, independent of the position, and the following modulation function:

$$\begin{cases} \varphi(t) = (\frac{t}{t_1})^2 & \text{if } 0 \leq t < t_1 \\ \varphi(t) = 1 & \text{if } t_1 \leq t < t_2 \\ \varphi(t) = e^{-a(t-t_2)} & \text{if } t \geq t_2 \end{cases} \quad (46)$$

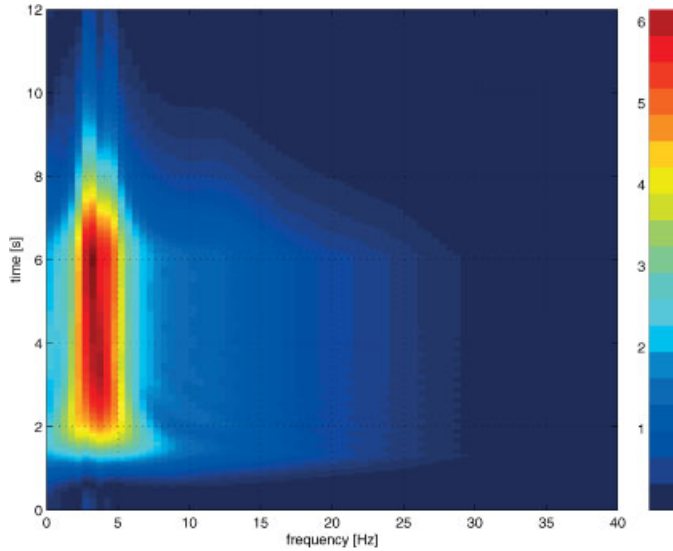


Figure 6. Square rooted, normalized instantaneous power spectral density of horizontal differential displacements on top of the buildings for $L_{\text{cor}} = R$.

with $a = 3.3$, $t_1 = 0.2$ s and $t_2 = 1$ s. The correlation length of the incident waves can be defined by

$$L_{\text{cor}}^{(j)}(\mathbf{x}, \omega) = \int_0^{+\infty} \gamma_{jj}(\mathbf{x} + u\mathbf{e}_j, \mathbf{x}, \omega) du \quad (47)$$

which leads to

$$L_{\text{cor}}(\omega) = \frac{\sqrt{\pi}}{2} \times \frac{c^s}{\eta\omega}, \quad j = 1, 2, 3 \quad (48)$$

for the model of Luco and Wong [5], thus showing in the mean time a possible definition of the coherency parameter η .

Figure 6 displays the instantaneous power spectral density of horizontal differential displacements ($\mathbf{e} = \mathbf{e}_1$) on top of the buildings, normalized by S_{0j} , for $L_{\text{cor}} = R$ where R is the radius of foundations. Much of the power is concentrated on the first eigenfrequency of the buildings (at about 4 Hz with coupling, compared to 10.8 Hz without coupling). Figure 7 displays the standard deviation $\sigma_{\Delta}(t)$ of horizontal differential displacements on top of the buildings as well, normalized by ε . It is seen that σ_{Δ} increases when the correlation length of the incident seismic waves decreases, in agreement with physical intuition. We also conclude from this plot that differential displacements are very much likely to exceed the gap ε between adjacent buildings thus leading to pounding (of course non-linear contact due to the shocks between the structures when $\Delta(t)$ exceeds ε has not been considered here). The large shift observed between fundamental eigenfrequencies of the buildings without and with coupling is due to the important contrast of stiffness between the soil and the structures. Clearly neglecting soil–structure interactions is not an acceptable hypothesis in the present case. A

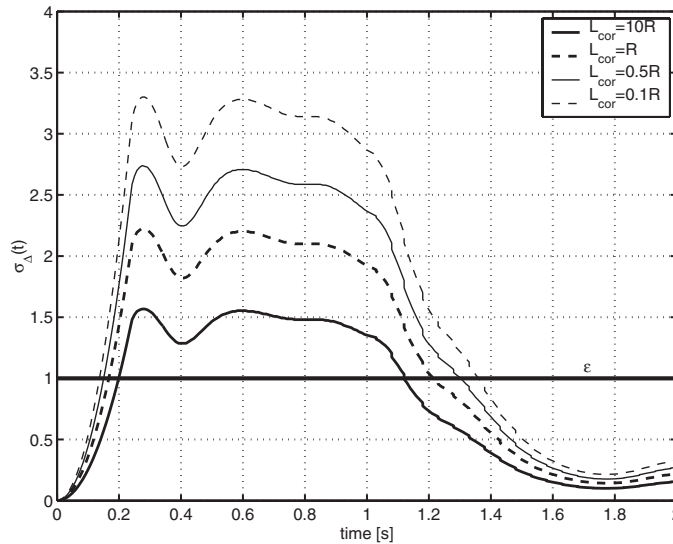


Figure 7. Normalized standard deviation of horizontal differential displacements on top of the buildings for $L_{\text{cor}} = 10R$, $L_{\text{cor}} = R$, $L_{\text{cor}} = R/2$ and $L_{\text{cor}} = R/10$.

simple calculation of the maximum displacement on top of the buildings associated to their first eigenmode shows that it is of the order of $\varepsilon/100$, and then differential displacements may never exceed the fixed gap between the structures if soil–structure interaction is ignored.

7. CONCLUSIONS

An indirect integral representation of displacements of a structure under incident seismic waves has been presented. It takes into account the linear soil–structure interaction effects. It is based on the knowledge of the fundamental solution (Green function) for the overall domain constituted by the structure and the soil when a unit load is applied at the location where structural response is observed. This Green function is computed by classical dynamic substructuring techniques and boundary integral equations. The influence of spatial variability of the incident waves can be evaluated by the standard spectral analysis theory. Their non-stationarity is addressed by introducing a deterministic modulation function of the earthquake motion process. The proposed procedure has been applied to various practical cases, and notably an investigation of pounding effect between two adjacent, identical structures resting on a thin soft soil layer which has shown the need to account for soil–structure interaction. Although the results obtained so far are applicable to any kind of grounded or superficial foundations, only the latter have been considered in the applications. This is because coherency functions for seismic waves are usually known only for surface motions. Analyses of soil–structure interaction for deep foundations by the present procedure require further investigations of the in-depth coherencies of earthquake motions. However, the proposed general model remains valid as long as the linear hypothesis is not violated. The overall modelling strategy retained does not extend to a non-linear behaviour of the soil since in this case neither the BEM nor the

spectral theory of linear filtering of stochastic processes are applicable. An alternative model and different mathematical and numerical tools shall be used for that purpose. Therefore the conclusions of this paper may not stand for a non-linear soil.

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